

Incentivizing Black-Box Model Sharing with Fair Rewards and Payoffs

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Privacy Preserving
Share only predictions,
no gradients or data.



Fair & Self-Interested
Rewards and payoffs are
proportional to contribution.



Better Together
All parties improve models
while being compensated.

1 MOTIVATION

Agents are self-interested.
Without proper incentives,
free-riding occurs.

Our goal:

A mechanism that is

✓ Fair

Rewards proportional to
contribution

✓ ϵ -Individual Rational

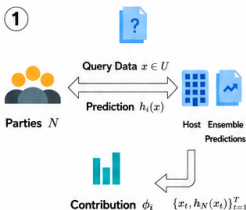
All parties are better off
participating

✓ Weakly Efficient

At least one party receives
the full ensemble reward

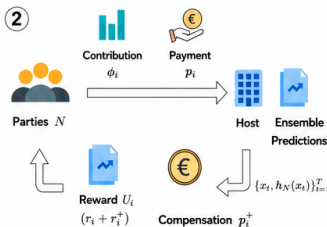
2 FRAMEWORK: A UNIFIED 2-STAGE MECHANISM

Stage 1: Contribution Evaluation (Weighted-Ensemble Game)



The host aggregates predictions to compute
each party's fair contribution ϕ_i (average
ensemble weight).

Stage 2: Joint Reward & Payment Allocation (Fair-Replication Game)



Parties receive data rewards (improve models) and
monetary payoffs (compensate budget) in a fair and
individually rational manner.

3 OUR TECHNICAL CONTRIBUTIONS

1



Fair Valuation via Weighted-Ensemble Game

- Value of a coalition C : $V(C) = \frac{1}{T} \sum_{x \in U} \sum_{i \in C} \beta_{i,x}$
- Shapley value (contribution): $\phi_i = V(\{i\}) = \frac{1}{T} \sum_{x \in U} \beta_{i,x}$

Theoretical guarantee (generalization bound)

$$L_D(h_N) \leq \sum_{i=1}^n \left(\phi_i + \sqrt{\frac{2 \log(2/\delta)}{T}} \right) L_D(h_i) + \Sigma_N$$

A larger ϕ_i means a model contributes more to lowering ensemble error.

2



Joint Allocation via Fair-Replication Game

We combine WEG (data value) with FRG (money-data exchange).
Let $V_N = \sum_i \phi_i$ and total payment pool $P = \sum_j p_j$.

Main allocation (combined Shapley value)

$$\begin{aligned} u_i &= r_i + r_i^+ + p_i^+ - p_i \\ r_i &= \phi_i & r_i^+ &= \frac{V_N \cdot p_i}{V_N - \phi_i} & p_i^+ &= \sum_{j \neq i} \frac{\phi_j \cdot p_j}{V_N - \phi_j} & - p_i \\ \text{Baseline reward} & & \text{Additional reward} & & \text{Compensation from others} & & \text{Payment} \end{aligned}$$

4 INCENTIVE GUARANTEES



Fairness

Rewards and net payoffs are proportional to contribution ϕ_i .



ϵ -Individual Rationality

Participation guarantees (almost) no worse performance
and additional reward is worth the payment.



Weak Efficiency

At least one party receives the full ensemble V_N .

ϵ -IR Guarantee (Virtual Regret Bound)

With high probability, $\forall i \in N$, $L_D(h'_i) - \epsilon \leq L_D(h_i)$ where

$$\epsilon_i = A \left(\frac{\alpha_i^2}{T_i} + \frac{(1 - \alpha_i)^2}{m_i} \right)^{\frac{1}{2}} + \alpha_i B + d_{H,DH}(D_i, D)$$

T_i : reward size (samples)

m_i : local data size

α_i : mixing weight

$L_D(h_N)$: ensemble error

$d_{H,DH}(D_i, D)$: distribution divergence



Key Insight

$\epsilon_i(\alpha_i)$ is minimized at an optimal α_i^* , yielding the strongest
 ϵ -IR guarantee and the largest empirical improvement.

In practice, the strict IR ($L_D(h'_i) \leq L_D(h_i)$) is consistently achieved.



Payoff Balance

$$\sum_i (p_i^+ - p_i) = 0$$



Dummy Payment

No contribution
 $\Rightarrow u_i = 0$



Semi-Symmetry

Equal marginal
contribution $\Rightarrow$
equal utility



Strict Monotonicity

Higher payments by
others $\Rightarrow$ higher
utility for all

TAKEAWAY

Our unified mechanism fairly exchanges money for data,
ensuring privacy, fairness, and rational participation—
empirically leading to better models for everyone.

APPLICATIONS



Healthcare
Institutions



Financial
Organizations



IoT & Edge
Devices



Research
Consortia